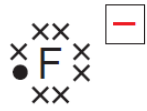
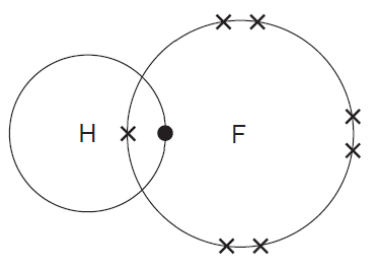
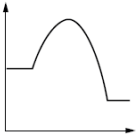
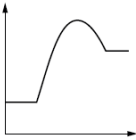
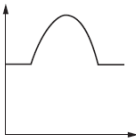


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		award (1) each for charge and electronic structure  fluoride ion (2,8)		2		2		
		(ii)		$2\text{Na} + \text{F}_2 \longrightarrow 2 \text{NaF}$ award (2) for correct equation if incorrect award (1) for correct formula of product		2		2		
	(b)			 award (1) for shared pair award (1) for fluorine octet		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		award (1) for any of following 436 + 154 add the bond energies 436 and 154 add H—H and F—F		1		1	1	
		(ii)		1130		1		1	1	
		(iii)		Energy  Reaction pathway ☒ Energy  Reaction pathway ☐ Energy  Reaction pathway ☐		1		1		
				Question 7 total	0	9	0	9	2	0